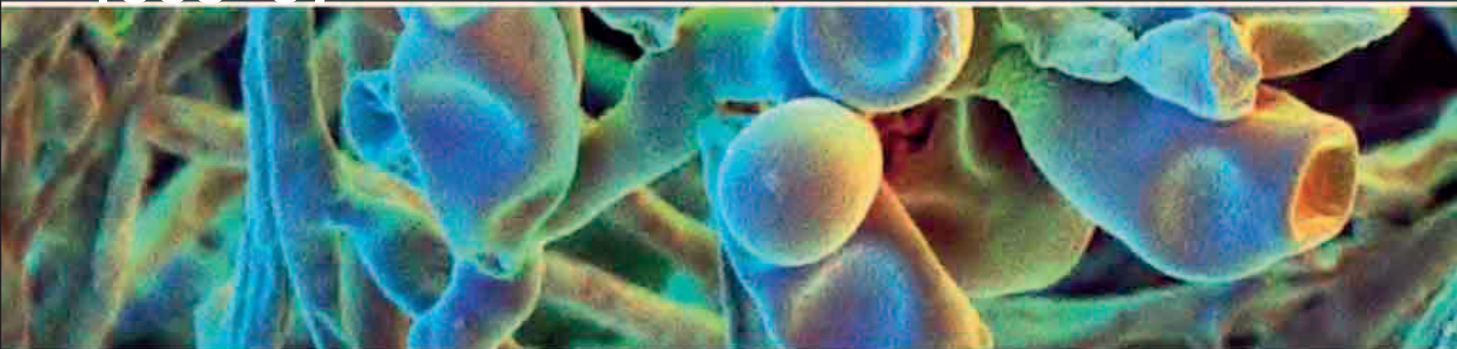




Dutchman Antony van Leeuwenhoek was one of the first observers of microscopic organisms such as these mold spore capsules.

SOME 20 YEARS AFTER HE MADE HIS FIRST OBSERVATIONS

of miniature life (see 1675–84), Dutch microscopist **Antony van Leeuwenhoek** published a compilation of his work, *Arcana Naturae* (*Secrets of Nature*) in 1695. As well as describing and illustrating a range of biological curiosities—from tadpoles to red blood cells—the book contained descriptions of the techniques Leeuwenhoek had used to carry out his studies. Many of these, including his microscope, were his own inventions.

In the same year, English theologian and mathematician **William Whiston** (1667–1752) published his *New Theory of the Earth*, which was a combination of religious and scientific thought. He supported the idea of divine creation, and his work

was praised by many, including Isaac Newton. Whiston suggested that the global catastrophe of the Biblical flood had been caused by a comet. He would succeed Newton as the third Lucasian

Professor of Mathematics at Cambridge University in England.

In 1697, many decades before the discovery of oxygen, German chemist **Georg Stahl** (1660–1734), proposed a **theory to explain combustion**. He suggested that metals and minerals contained two components—one being the **calx** (ashy residue), and the other being a substance called



main screw

body plate

Simple microscope

In his 1695 study *Arcana Naturae*, Antony von Leeuwenhoek explained the use of the microscope he had designed himself.

phlogiston, which was given off when something burned. Stahl thought that the amount of phlogiston varied: there was a great deal in coal, which diminished to ashes during combustion, but very little in iron, which did little more than rust. The phlogiston theory had its roots with Stahl's mentor, German alchemist Johann Becher, who conceived phlogiston as *terra pinguis* (oily matter), one of the classical elements. Later, the French chemist Antoine Lavoisier argued that **combustion** happened by **oxidation**: reaction of the substance with oxygen in the air. Stahl's idea of the calx was equivalent to the modern idea of oxide.

In 1697, Swiss mathematician **Johann Bernoulli**, prompted by a dispute with his brother (who was often his bitter rival), **solved a trajectory problem**. He described the path followed by a particle moving under

gravity. By studying the rates of movement along this curve, Bernoulli's work had important implications for the development of calculus: the mathematics of infinitesimal changes.

Also in 1697, English explorer **William Dampier** (1651–1715) published an account of his first voyage, *A New Voyage Around the World*—containing descriptions of the Americas and East Indies. The British Admiralty granted him the command for another trip, and Dampier eventually circumnavigated the globe three times. His **work on navigation influenced explorer James Cook**, while his studies of natural history would be used by biologists such as Alexander von Humboldt and Charles Darwin (see 1859).



The first dissection of a chimpanzee, in 1698, revealed a humanlike brain.

ISAAC NEWTON HAD ALREADY ESTABLISHED

that sound moves as longitudinal compression waves, not by transverse waves (oscillation at right angles to the direction of travel), as previously thought. In 1698, he went on to calculate the **speed of sound in air**, which he determined to be 979 ft (298 m) per second. (The modern value is 1,125 ft per second.)

Dutch astronomer Christiaan Huygens died in 1695, but his final book, *Cosmotheoreos*, appeared in 1698. He had delayed publication because he feared offending religious sensibilities—he had conjectured upon the possibility that life existed on other planets with habitable conditions.

British physician **Edward Tyson** (1650–1708) was governor of the Bethlem Hospital for psychiatric patients, in London. He routinely performed

1,125 FEET PER SECOND THE SPEED OF SOUND

JOHANN BERNOULLI (1667–1748)

Bernoulli was born into a prominent mathematical family and had professorships in Groningen, the Netherlands, and Basel, Switzerland. His work included studying the mathematical trajectories of curves and investigating the reflection and refraction of light. Together with his brother Jacob, he helped Newton and Leibniz develop calculus.



1695 Antony van Leeuwenhoek publishes *Arcana Naturae*

1695 William Whiston publishes *New Theory of the Earth*

1697 Johann Bernoulli solves a trajectory problem

1697 Georg Stahl proposes the phlogiston theory of combustion

1697 William Dampier publishes an account of his first world voyage

1698 English mathematician Isaac Newton measures the speed of sound